

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

GRAVITATION

LECTURE

3

PHYSICS

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Topics To Be Covered

- 1 Revision ✓
- 2 Escape Velocity ✓
- 3 Orbital Velocity ✓
- 4 Kepler Law of Planetary Motion ✓ 3

Recap From 1st Lecture

* Universal Law of Gravitation

$$F = G \frac{m_1 m_2}{r^2} \quad (m_1) \text{---} r \text{---} (m_2)$$

$$G = 6.67 \times 10^{-11} \frac{\text{Nm}^2}{\text{kg}^2}$$

Vector form of Gravitational Law

$$F = \frac{G m_1 m_2}{(r_2 - r_1)^3} (\vec{r}_2 - \vec{r}_1) \quad \text{JKBOSE}$$

Acceleration due to Gravity



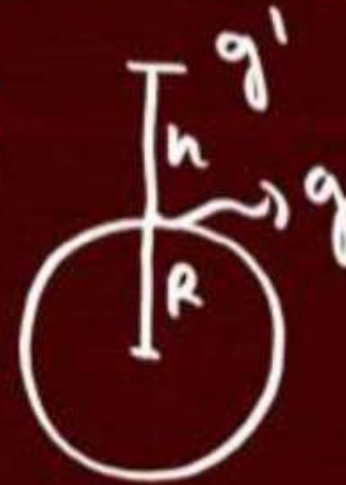
$$g = \frac{G M}{r^2}$$

$$g = G \frac{4}{3} \pi \rho r$$

$$g = -10 \text{ m/s}^2$$

$$g = -9.8 \text{ m/s}^2$$

Variation of g with height

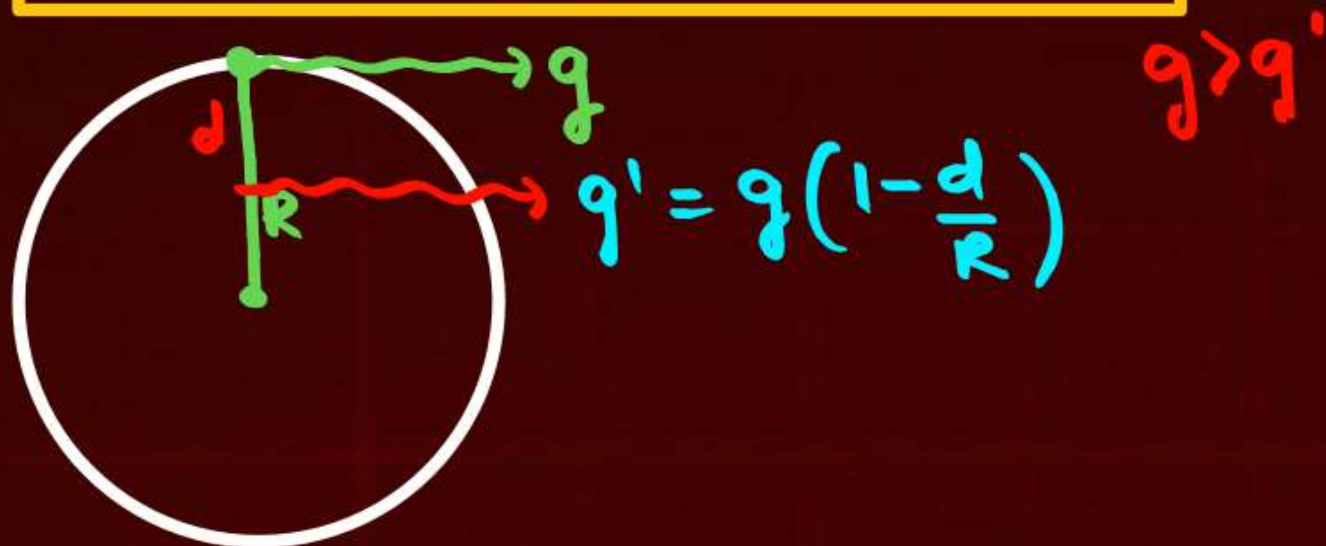


$$g' = g \left(1 - \frac{2h}{R}\right)$$

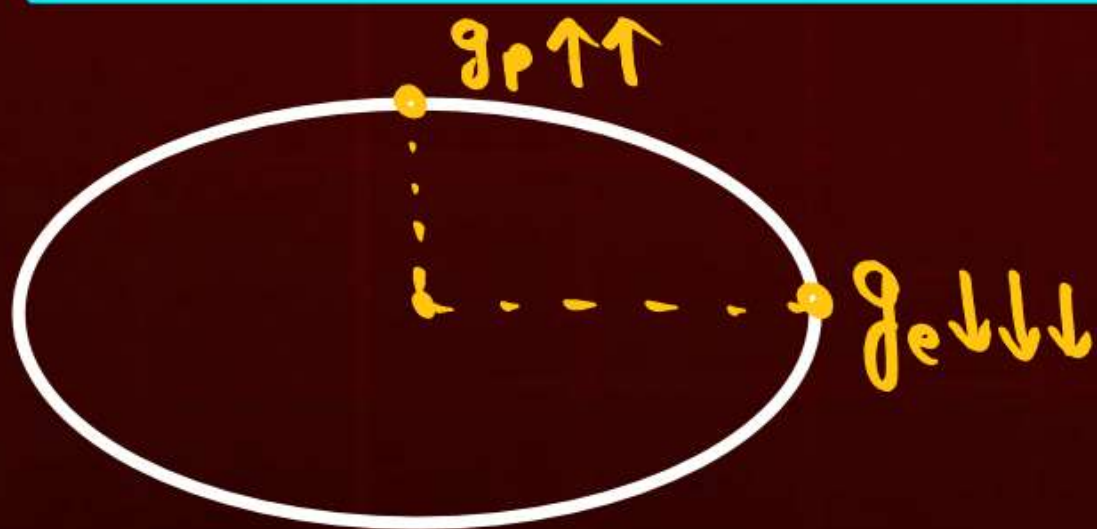
$$g > g' \checkmark$$

Recap From 2nd Lecture

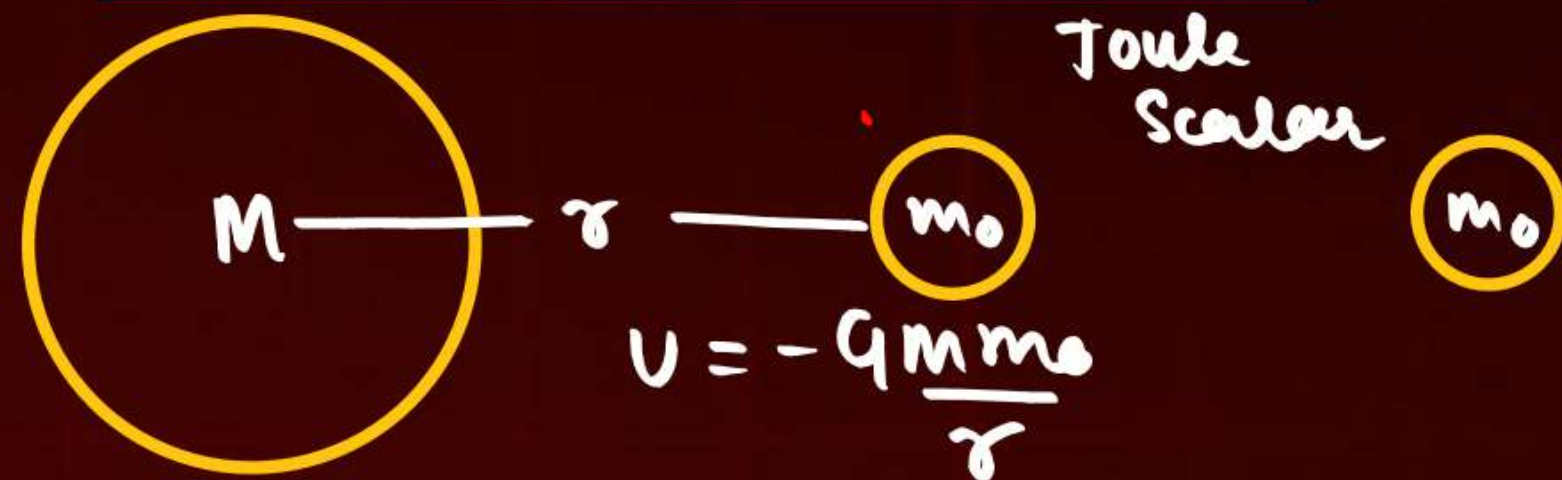
Variation of g with depth



Variation of g due to Shape of Earth



Gravitation Potential Energy



Gravitational potential

Scalar

$$\checkmark V = \frac{U}{m_0} \quad \frac{\text{J}}{\text{kg}}$$

$$\checkmark V = -G \frac{M}{r}$$

Escape Velocity

throw

The velocity with which you can project a object so that it leave the earth gravitational field.

$$V_e = \sqrt{\frac{2Gm}{r}} \text{ (Derivation ✓)}$$

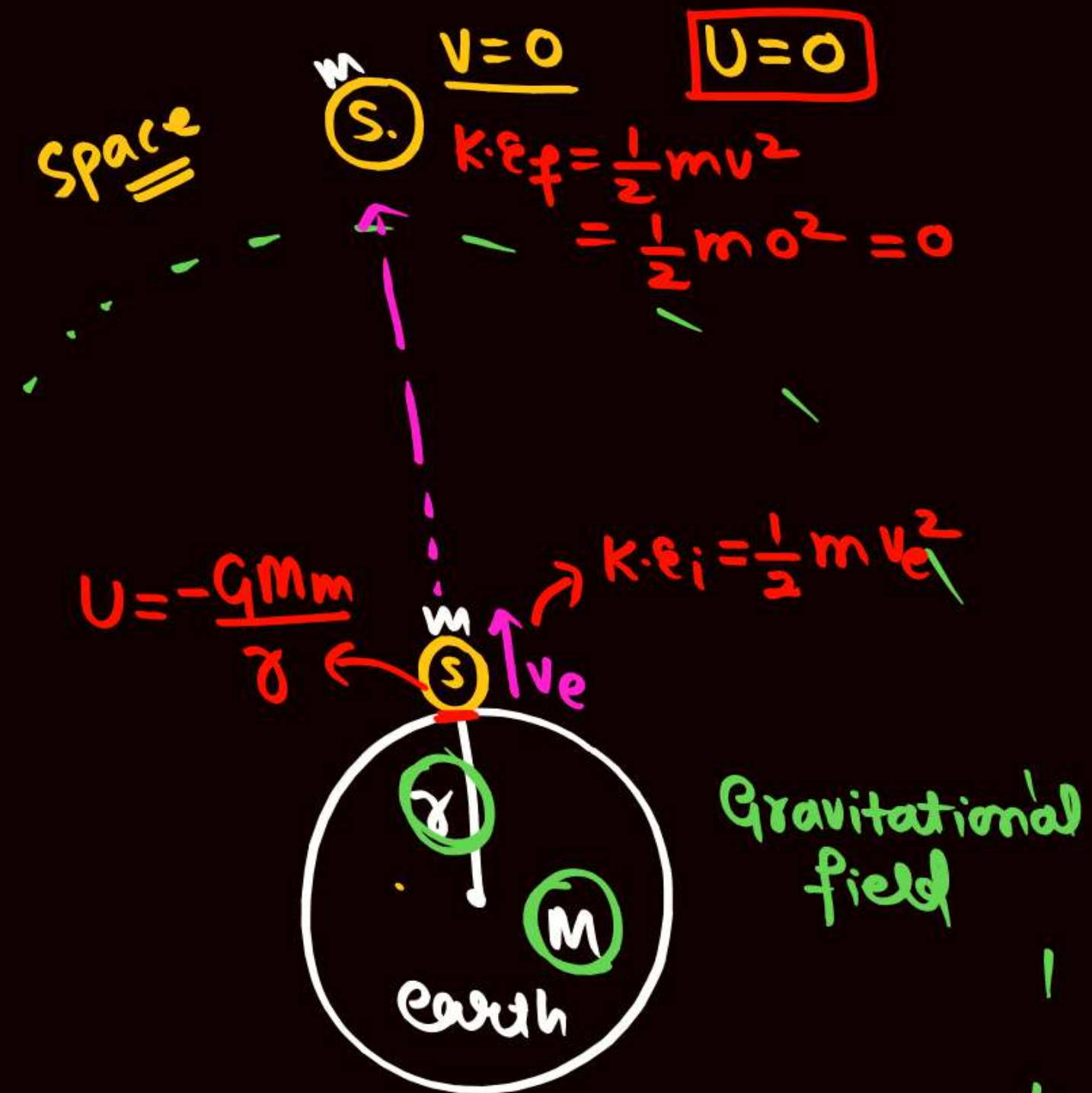
$$V_e = \sqrt{2rg}$$

$$V_e = 11.2 \text{ km/sec}$$

$$\text{unit} = \frac{\text{m}}{\text{Sec}}$$



Derivation



Law of Conservation of Energy

$$T.E_i = T.E_f$$

$$\underline{K.E_i} + P.E_i = K.E_f + P.E_f$$

$$\frac{1}{2}mv_e^2 + \underbrace{-\frac{GMm}{r}} = 0 + 0$$

$$\frac{1}{2}mv_e^2 = \frac{GMm}{r}$$

$$\frac{1}{2}v_e^2 = \frac{Gm}{r}$$

$$v_e^2 = \frac{2Gm}{r}$$

$$v_e = \sqrt{\frac{2Gm}{r}}$$

2nd formula Escape Velocity

$$V_e = \sqrt{2 \frac{Gm}{r}} \quad r g \quad g = \frac{Gm}{r^2}$$

$$V_e = \sqrt{2 r g}$$

$$g = \frac{Gm}{r \times r}$$

$$r g = \frac{Gm}{r}$$

Note :- V -Escape do not depend upto the mass of object which is projected.

Numerical

Q Calculate the escape velocity of earth?

(a) 11.2 km/sec ✓

(b) 12.2 km/sec

(c) 15.2 km/sec

(d) 18.2 km/sec

$v_e = \sqrt{2rg}$

$r = 6400 \text{ km}$
 $r = 6.4 \times 10^6 \text{ m}$
 $g = 9.8/10$

$$v_e = \sqrt{2 \times 6.4 \times 10^6 \times 10}$$

$$v_e = 11,200 \text{ m/sec}$$

$$v_e = \frac{11200}{1000} \text{ km/sec}$$

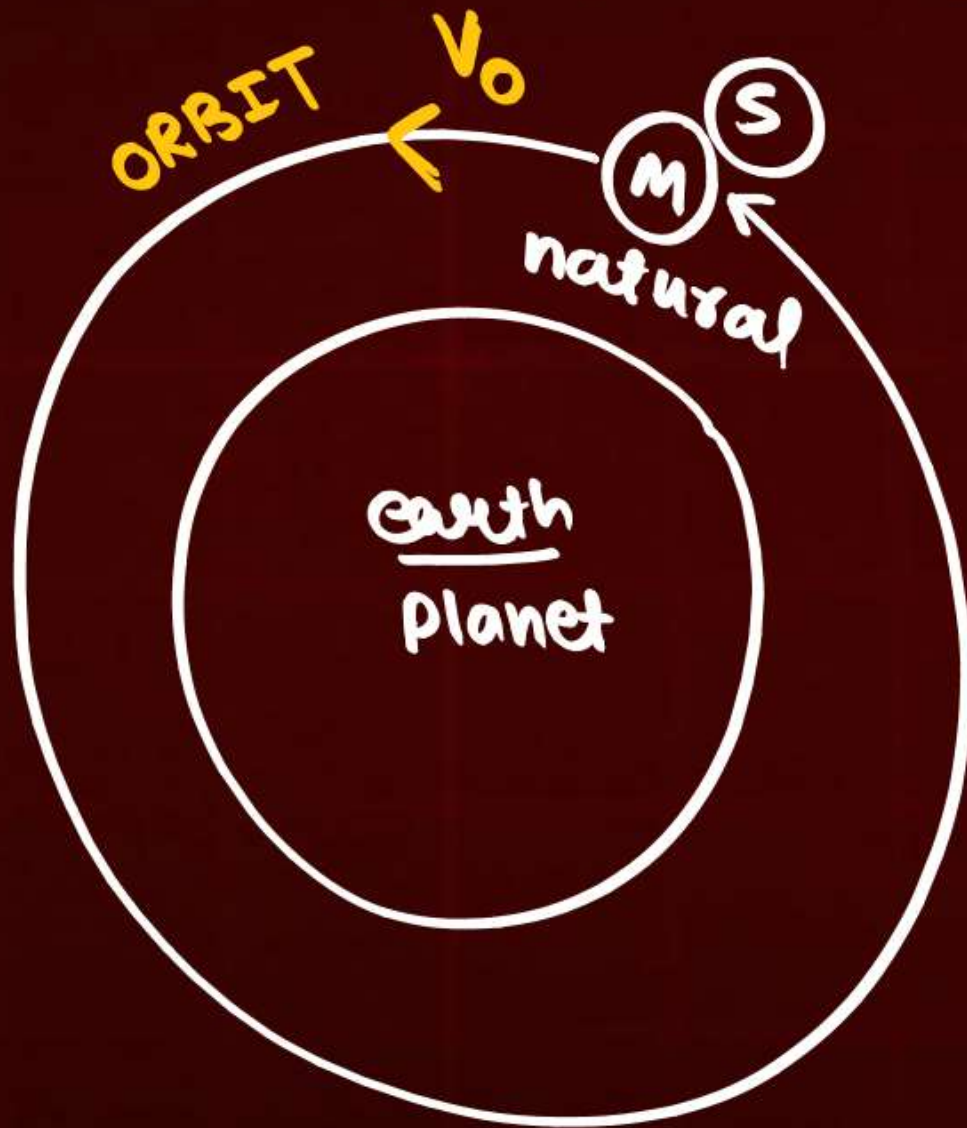
$$v_e = 11.2 \text{ km/sec}$$

Orbital Velocity

$$V_o$$

Satellite ✓

The heavy body revolving around another heavy body (planet)



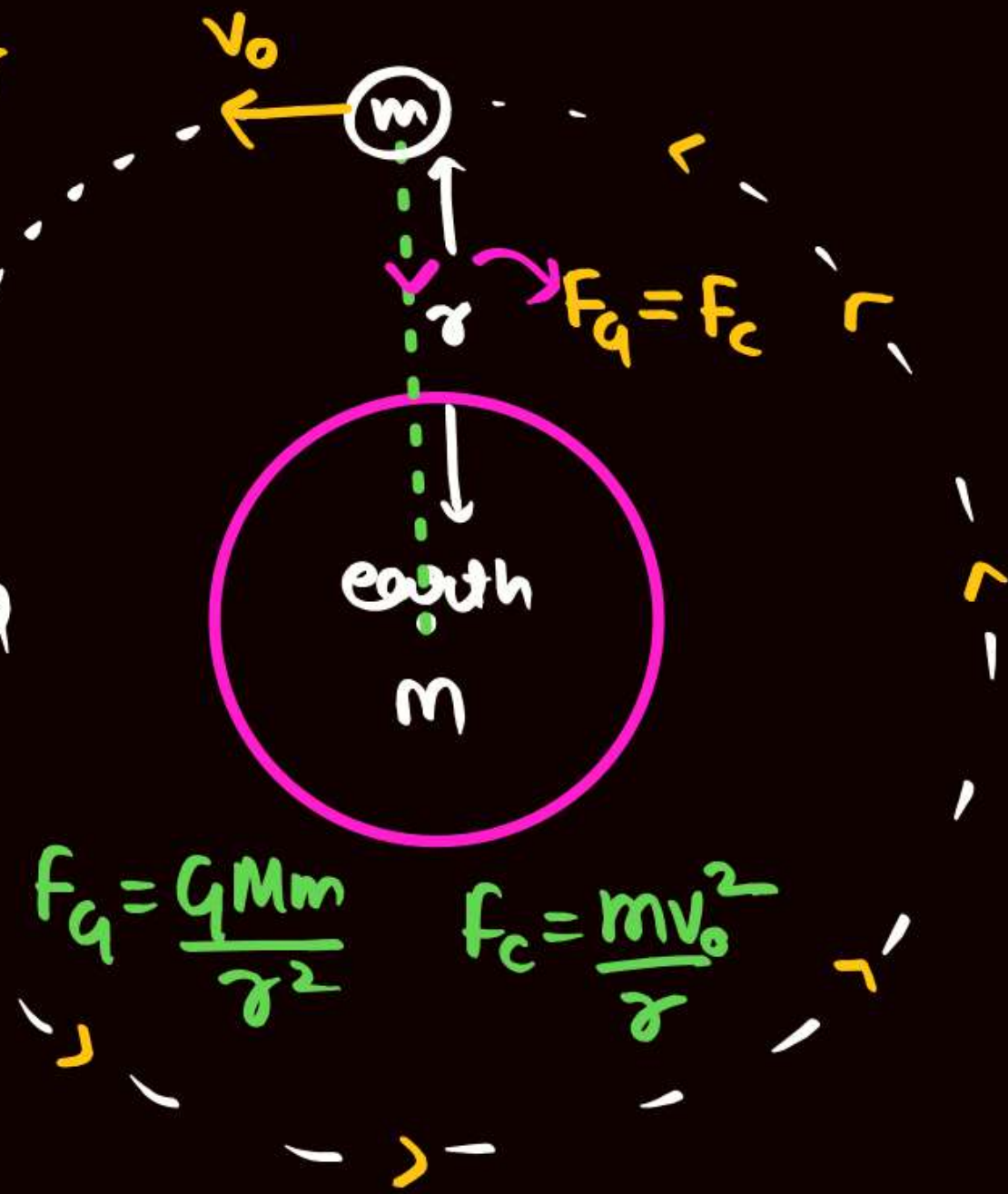
Orbital velocity

The velocity with which satellite revolve around the planet.

$$V_0 = m/s$$

$$V_0 = \sqrt{\frac{GM}{r}}$$

$$V_0 = \sqrt{rg}$$



$$F_g = \frac{GMm}{r^2}$$

$$F_c = \frac{mv_0^2}{r}$$

Derivation (V₀)

necessary F_c is provided by F_g

$$F_g = F_c$$

$$\frac{GMm}{r^2} = \frac{mv_0^2}{r}$$

$$\frac{GM}{r} = v_0^2$$

$$v_0^2 = \frac{GM}{r}$$

$$V_0 = \sqrt{\frac{GM}{r}} \quad \text{--- (1)}$$

Another formula of V_0

$$V_0 = \sqrt{\frac{Gm}{r}}$$

$$g = \frac{Gm}{r^2}$$

$$V_0 = \sqrt{rg}$$

$$g = \frac{Gm}{r \times r}$$

$$rg = \frac{Gm}{r}$$

2nd formula

Q Relationship b/w V_e and V_o

JKBOSE — 2014

2016

2017

2021

2022

Sol As we know that

$$V_e = \sqrt{2 \times \frac{GM}{r}}$$

$$V_o = \sqrt{\frac{GM}{r}}$$

$$V_e = \sqrt{2} \left(\sqrt{\frac{GM}{r}} \right)$$

$$V_e = \sqrt{2} V_o$$

8th
Math

Q Calculate the V_0 for earth (Numerical) - Jee 2019

(a) 11.2 km/sec $\checkmark e$

(b) 7.9 km/sec $\checkmark o$

(c) 8.1 km/sec

(d) 8.7 km/sec

Sol $V_e = \sqrt{2} V_0$ $V_0 = 7.9 \text{ km/sec}$

$11.2 = \sqrt{2} V_0$

$11.2 = V_0$

$\sqrt{2}$

$\frac{11.2}{1.4} = V_0$

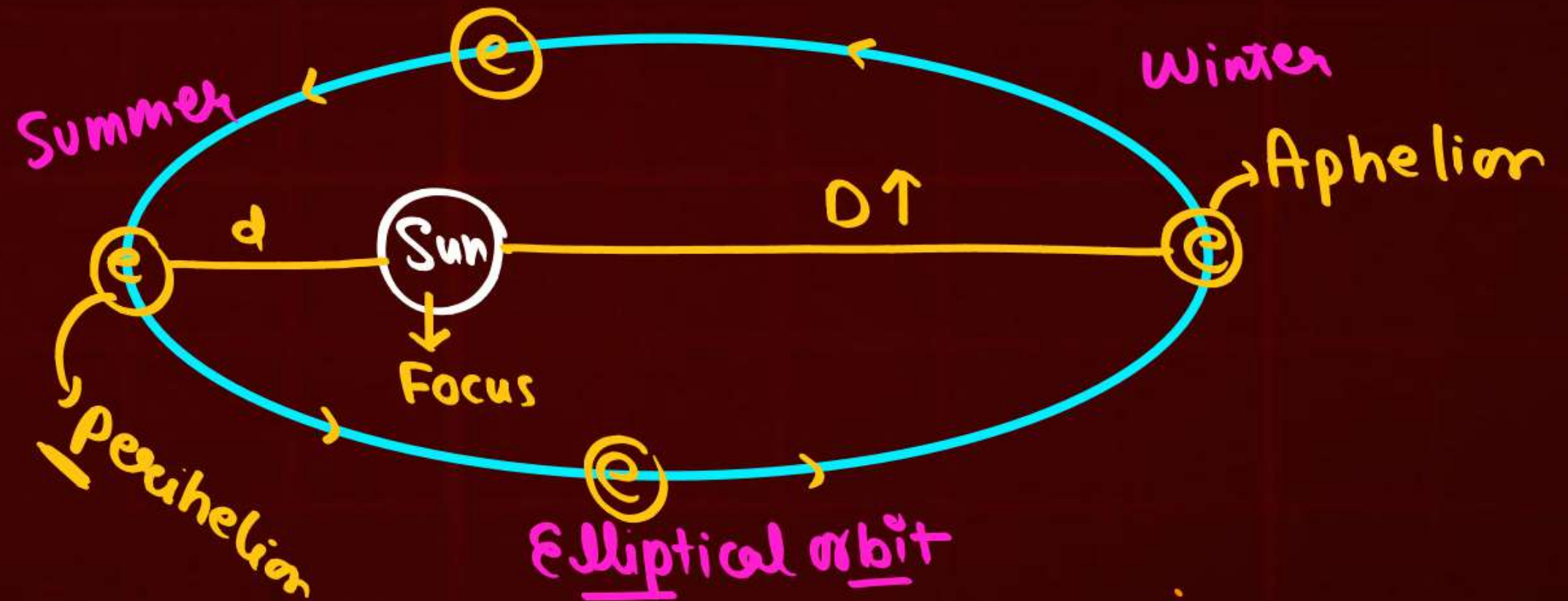
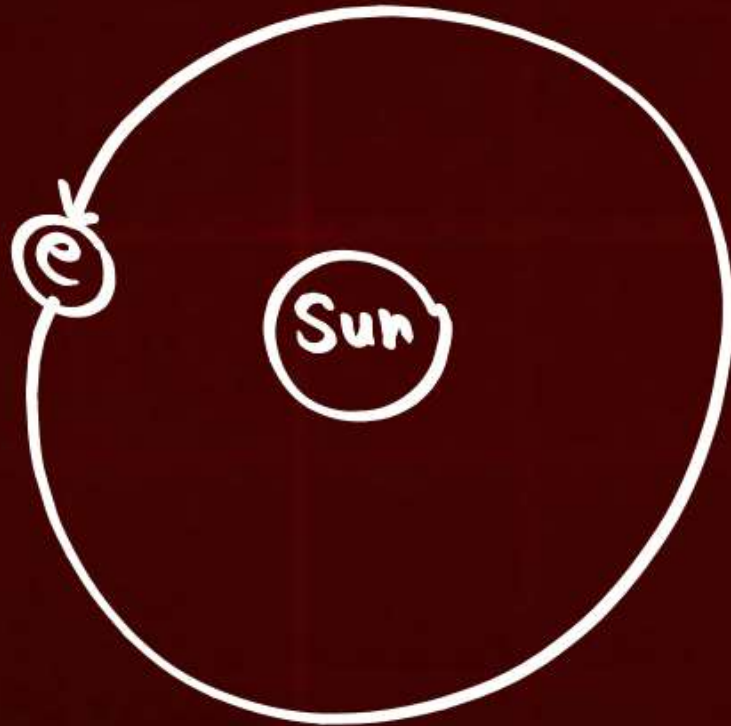
$\sqrt{2} = 1.4$

Keplar Law of Planetary Motion

① Law of Orbit

It state that plant^e moves in elliptical orbit around Sun

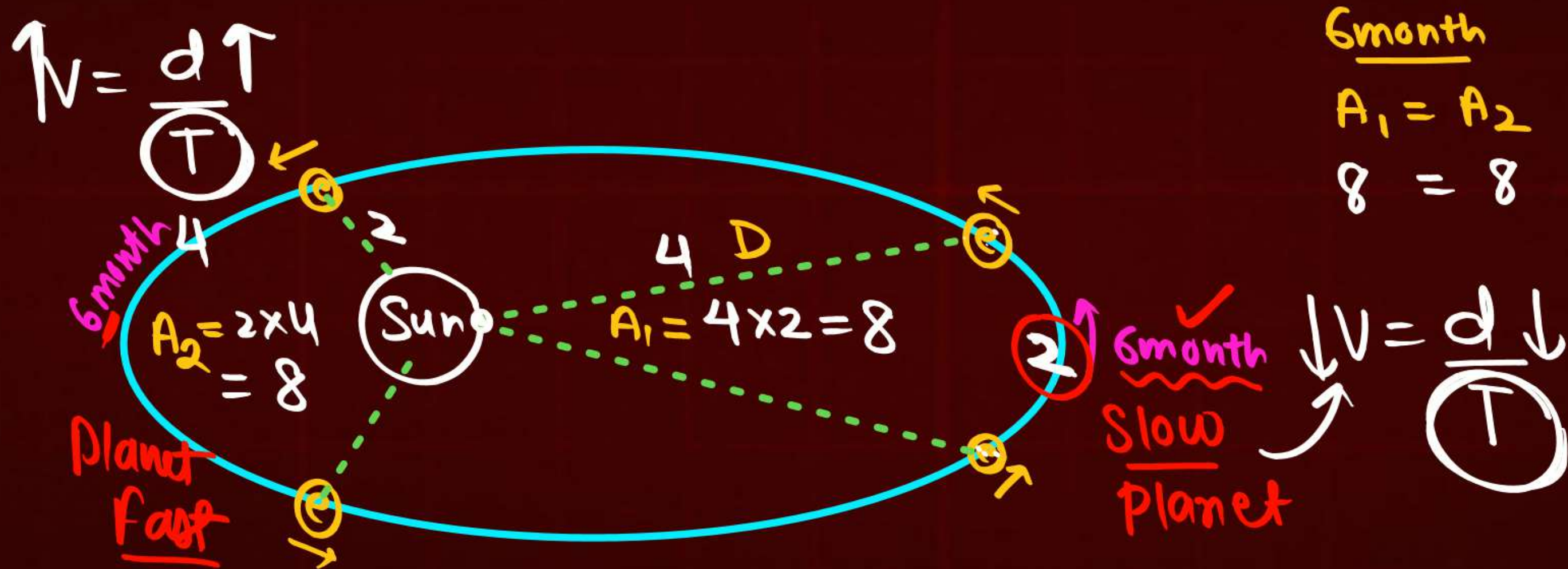
X circular orbit



Keplar Law of Planetary Motion

Law of Area

The Planet Sweep out equal area in equal interval of time

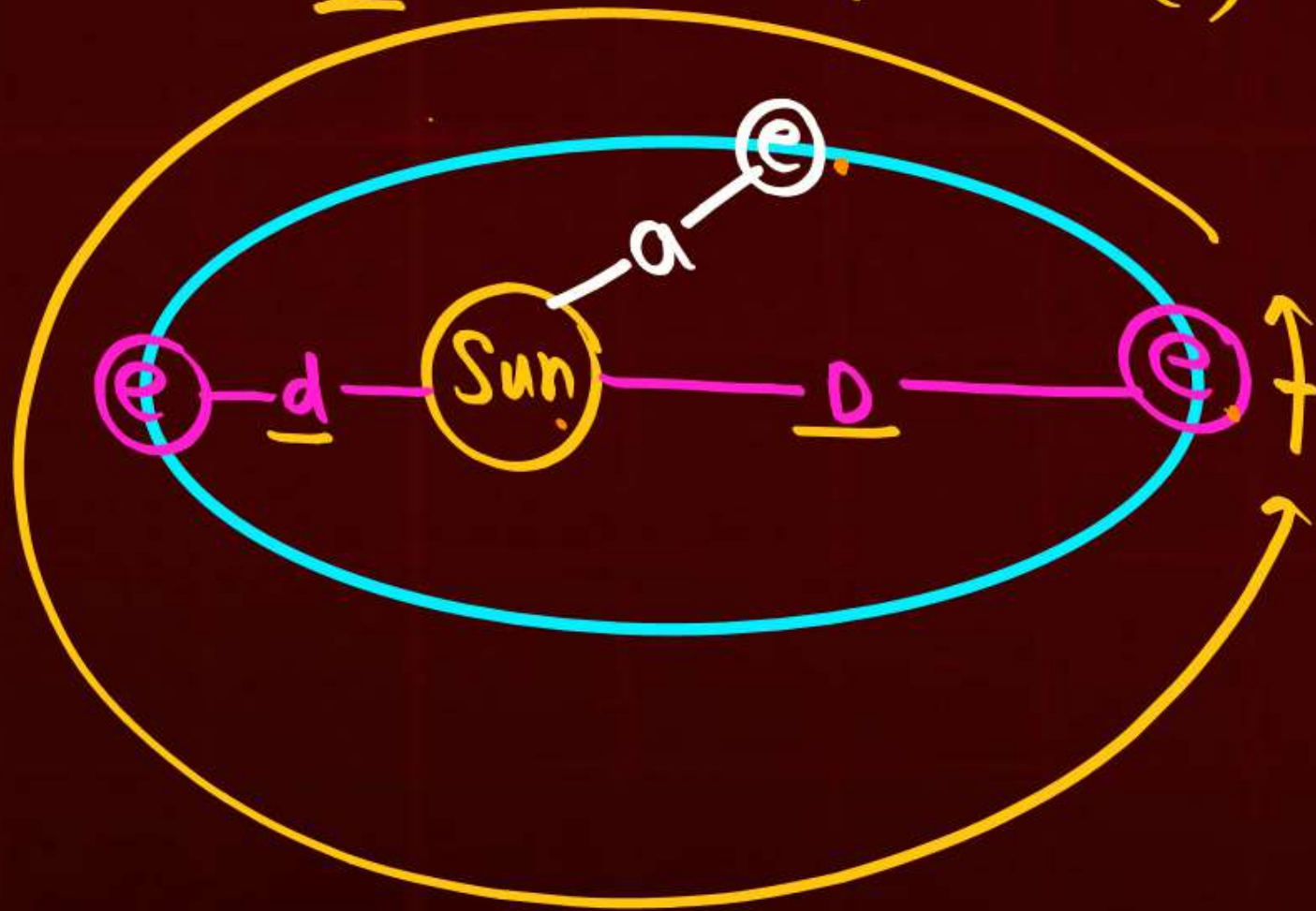


Keplar Law of Planetary Motion

Law of time period

The square of time (T) is directly proportional to the cube of semi major axis.

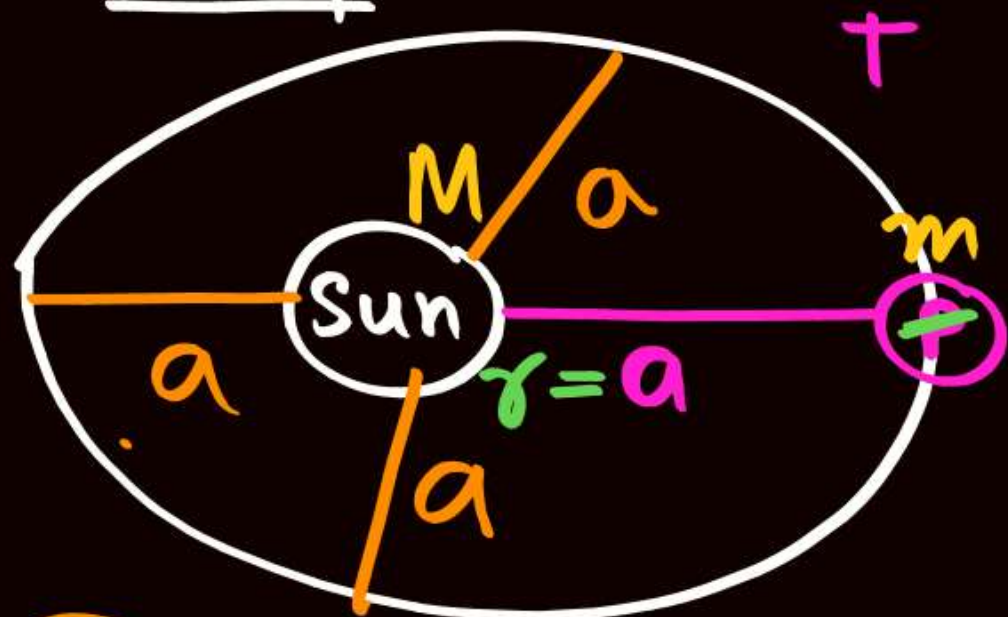
$$a = \frac{D+d}{2} \quad \text{Time period (T)}$$



$$T^2 \propto a^3$$

average distance b/w
Sun and planet

Proof



Circular orbit

$$V = \frac{d}{T} = 2\pi r$$

$$V = \frac{2\pi r}{T}$$

$$V = \frac{2\pi a}{T}$$

$$\# F_c = F_g$$

$$\cancel{m}v^2 = \cancel{G}M\cancel{m}$$

$$a = r^2$$

$$v^2 = \frac{Gm}{a}$$

$$\left(\frac{2\pi a}{T}\right)^2 = \frac{Gm}{a}$$

$$\frac{4\pi^2 a^2}{T^2} = \frac{Gm}{a}$$

$$4\pi^2 a^2 a = GmT^2$$

$$4\pi^2 a^3 = GmT^2$$

$$GmT^2 = 4\pi^2 a^3$$

$$T^2 = \frac{4\pi^2}{Gm} a^3$$

$$T^2 = C a^3$$

$$T^2 \propto a^3$$

$$T^2 \propto a^3$$

Chalo Solve krte hai?

✓ H/W

Calculate the orbital velocity of an object revolving around earth?

Ye Duniya Chahay
Tumhe Jo Bhi Banne Ka
Mauka De Sub Se Pehle
Tum Rahem Dil Banna

Thank
You

